

Vaibhav Parasa

Data Scientist & ML Engineer · Agentic AI & LLM Systems

San Francisco, CA (open to relocation) · 612-229-9239 · vaibhav.v.parasa@gmail.com · linkedin.com/in/vaibhavparasa · github.com/starkparasa

EXPERIENCE

Lead Data Analyst / ML Engineer · HorizonPlus Technologies

Remote | Dec 2025 – Present

- Lead a client-facing agentic AI implementation, owning the architecture and workflows that automate processes previously handled manually.
- Integrated LLM and generative AI tooling into data workflows with LangChain, cutting manual effort in code generation and pipeline maintenance.
- Overhauled ETL/ELT pipelines across Databricks and Snowflake, expanding KPI coverage across the client's data model.
- Direct a team of data engineers and own Tableau reporting as primary analytical contact for client stakeholders.

Economic Data Analyst · State of Florida – AHCA

Tallahassee, FL | Feb 2025 – Jan 2026

- Produced financial analysis reports for healthcare licensure applications across 500+ organizations. Each required independent judgment, since no two applicants shared a financial structure, compliance history, or operational standing, so the work could not be templated.
- Built billing irregularity, budget variance, and branch audit KPIs in SQL and Python that flagged non-compliant organizations for state audit.
- Built pipelines from state databases through SharePoint into Tableau, making budget utilization visible to policy stakeholders, and identified organizations with the financial standing to serve designated areas of need, informing state approvals of new healthcare units.

Data Analyst · Aramark

Tallahassee, FL | Oct 2023 – Dec 2024

- Traced persistent food over-ordering to a flaw in the demand forecasting algorithm and was brought onto the redesign of that system.
- Built SQL ingestion pipelines for event-driven customer behavior data and the Python transformation logic behind the production forecasting model.
- Improved event resource planning accuracy 25% through statistical modeling, cutting food waste across campus operations, and presented the findings to senior management.

Business Intelligence Analyst · Accenture

Hyderabad, IN | Jun 2022 – Jun 2023

- Analysis was limited by having no direct line to the clients generating the data, so with the team established single-point client contacts. That change made accurate requirements, and the analytical work itself, possible.
- Built predictive models in Python and SQL forecasting market conditions and client demand, giving client teams a data-backed basis for strategic planning across full Agile sprints.
- Earned ServiceNow Administrator and Developer certifications, building custom platform modules in JavaScript, HTML, and CSS for a 10,000+ user enterprise system.

Software Developer – Backend · Infinos Technologies LLP

Hyderabad, IN | May 2021 – May 2022

- Built the database architecture and backend API layer for a mobile-controlled IoT cooling unit, letting users regulate temperature and access from anywhere over a stable connection, with multi-layer security and hashing designed in from the start rather than retrofitted.
- Optimized ESP-32-to-backend communication protocols, cutting device-to-interface latency 20% and reducing bandwidth demand on the unreliable connections that were usually the field constraint.
- Helped define data naming conventions and KPI structures for new application metrics, keeping the backend consistent from early in development.

PROJECTS

AI Travel Planner · FastAPI · agent loop · Ollama · MySQL · Docker · GitHub Actions

- Chat-driven itinerary generator built on an agent loop where the model decides whether to call weather and currency tools, folding results into generation rather than hardcoding the calls.
- Added an intent-classification step ahead of generation, routing each message to new trip, edit, question, or off topic. It removed nonsensical outputs, cut latency on turns needing no generation, and doubles as the prompt-injection guardrail.
- Chunked long itineraries into bounded LLM calls with capped conversation context, keeping local-model output reliable at any trip length; shipped with CI, Docker, and mocked-LLM tests.

Cart Abandonment Prediction · scikit-learn · Optuna · MLflow · LightGBM

- Classification pipeline on 100K e-commerce records: 60+ raw signals engineered into 30 interpretable behavioral features with target-leakage variables explicitly removed, then tuned with Optuna and tracked in an MLflow model registry.

SKILLS

Core	Python, SQL, Databricks, Snowflake, AWS, Spark, Tableau, Scikit-learn, Pandas
ML & AI	LangChain, Agentic AI, LLM Deployment, MLflow, Optuna, Predictive & Statistical Modeling, Forecasting
Infrastructure	ETL/ELT Pipelines, Data Warehousing, Cloud Architecture, Backend APIs, FastAPI, Docker, CI/CD
Platforms	ServiceNow (Admin & Developer Certified), JavaScript, HTML/CSS, IoT & ESP-32, Agile/SCRUM, KPI Design

EDUCATION

M.S. Data Science · Florida State University, Tallahassee FL

GPA 3.66 / 4.0 | Dec 2024

B.S. Computer Science · GITAM University, Hyderabad India

GPA 3.0 / 4.0 | May 2022